

and capacitors C1 through C6. When a bug passes through the grid, the high grid voltage causes a discharge though the bug, the grid voltage reduces and the stored voltage kills the bug instantly.

To ignite the UV lamp, a voltage tapping is provided from the voltage multiplier circuit. At switch-on,

around 700V ionises the gas in the tubelight and capacitor C7 discharges through the tube, lowering the resistance enough to sustain continual AC current flow. C7 then continues to act as the ballast.

Assemble the circuit on a general-purpose PCB and enclose in a small rectangular cabinet. Fix the UV lamp

and the assembled PCB inside the cabinet. Mount the high-voltage grid outside the cabinet on the front panel. Now your electric bug zapper is ready to use.

EFY warning. As the circuit works on a high voltage, don't touch the grid with bare hands to avoid electric shock, which can be dangerous! ●

WATER-LEVEL INDICATOR USING 7-SEGMENT DISPLAY

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This water-level indicator uses a 7-segment display, instead of LEDs, to indicate the water level (low, half and full) in the tank. Moreover, a buzzer is used to alert you of water overflowing from the tank. The circuit shows the water level by displaying L, H and F for low, half and full, respectively.

The circuit uses five sensors to sense the different water levels in the

tank. Sensor A is connected to the negative terminal (GND) of the power supply. The other four sensors (B through E) are connected to the inputs of NOT gate IC 7404. When there is a high voltage at the input pin of the NOT gate, it outputs a low voltage. Similarly, for a low voltage at the input pin of the NOT gate, it outputs a high voltage.

When the tank is empty, the input pins of IC 7404 are pulled high via a 1-mega-ohm resistor. So it outputs a

low voltage. As water starts filling the tank, a low voltage is available at the input pins of the gate and it outputs a high voltage.

When the water in the tank rises to touch the low level, there is a low voltage at input pin 5 of gate N3 and high output at pin 6. Pin 6 of the gate is connected to pin 10 of gate N9, so pin 10 also goes high. Now as both pins 9 and 10 of gate N9 are high, its output pin 8 also goes high. As a result, positive

supply is applied to DIS3 and it shows 'L' indicating low level of water in the tank.

Similarly, when water in the tank touches the half level, pins 4 and 5 of AND gate N8 become high. As a result, its output also goes high and DIS2 shows 'H' indicating half level of water in the tank. At this time, pin 9 of gate N9 also goes low via gate N4 and DIS3 stops glowing.

When the water tank becomes full, the voltage at pin 1 of gate N1 and pin 3 of gate N2 goes low. Output pin 3 of gate

